

Image Enhancement and Preprocessing

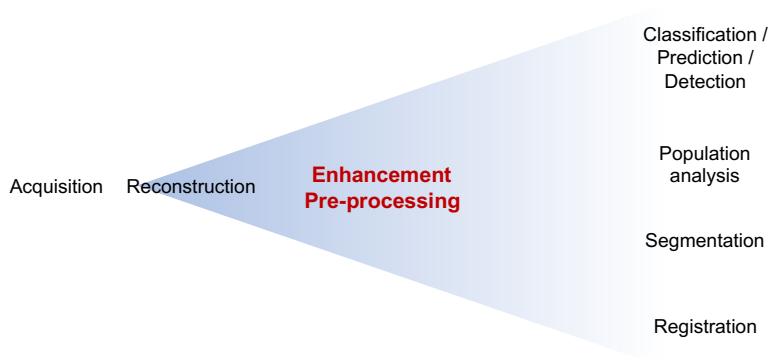
Ender Konukoglu

ETH Zürich

February 17 - 24, 2026

Section 1

Introduction



Enhancement and pre-processing tasks form initial steps for any type of analysis. Same applies to methods that use machine learning, probabilistic, variational or energy-based formulations.

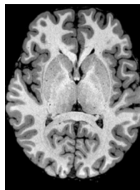
Image enhancement and pre-processing

Enhancement of four conditions

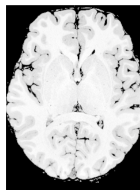
- Bad contrast
- Varying intensity statistics
- Noise
- Bias field
- Variation in pixel-size

Why enhancement?

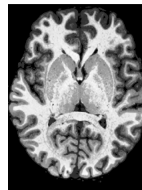
- simplifying interpretation
- better visualization
- normalization for further processing



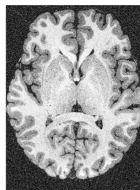
Original Image



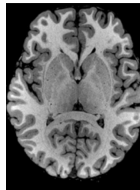
Bad contrast



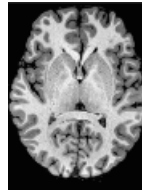
Intensity variation



Noise



Bias field



Different pixel-size

Outline

- Intensity normalization
- Noise suppression
- Exercise / Challenge
- Bias correction
- Variations in image and pixel-size
- Additional material
 - Contrast enhancement

Section 2

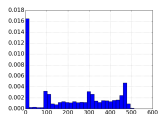
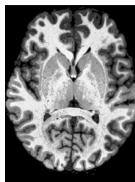
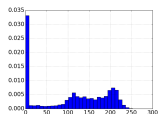
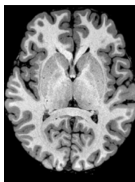
Intensity normalization

Outline

- Intensity normalization
 - Problem source
 - Basic normalizations
 - Histogram normalization with landmarks
- Noise suppression
- Exercise / Challenge
- Bias correction
- Variations in image and pixel-size
- Additional material
 - Contrast enhancement

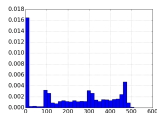
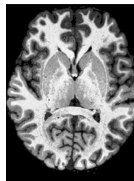
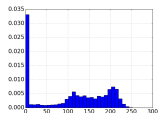
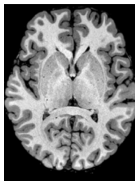
Problem source

- Intensity variations between images



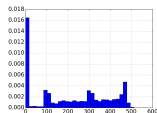
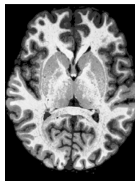
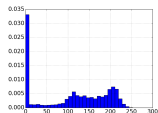
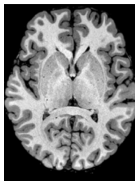
Problem source

- Intensity variations between images
- Differences in
 - Scanners
 - Protocols
 - Acquisition software
 - Reconstruction method
 - ...



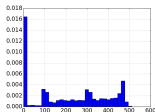
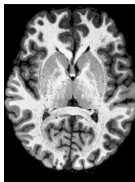
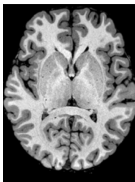
Problem source

- Intensity variations between images
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 - Scanners
 - Protocols
 - Acquisition software
 - Reconstruction method
 - ...
- Particularly important for MRI due to lack of absolute intensities



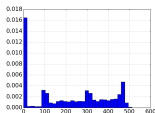
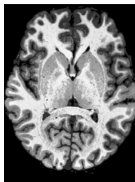
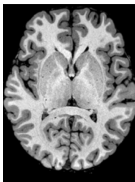
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- Particularly important for MRI due to lack of absolute intensities
- Less of a problem for modalities with absolute intensities, e.g. Hounsfield Unit in CT, however variations can still occur.

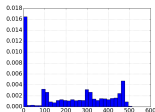
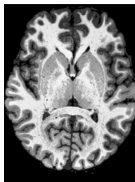
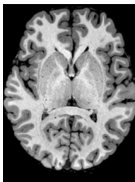


Problem source

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 - Scanners
 - Protocols
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- Less of a problem for modalities with absolute intensities, e.g. Hounsfield Unit in CT, however variations can still occur.
- Intensity variations can cause adverse effects for the rest of the processing steps

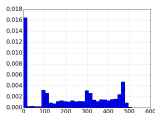
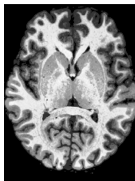
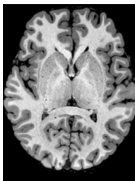


Problem source



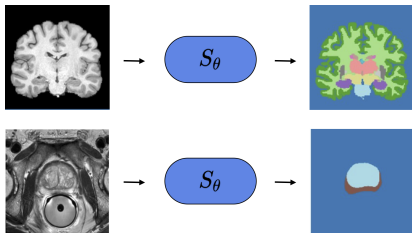
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- Intensity variations can cause adverse effects for the rest of the processing steps
- Particularly problematic for machine learning methods. Intensity differences between training and test samples is called domain shift. Think of images at night and day.

Problem source

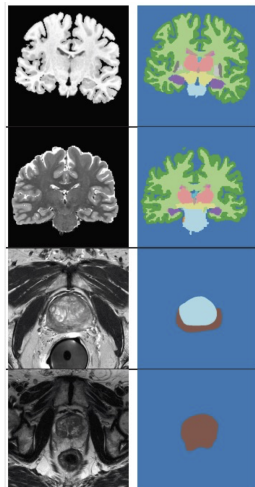
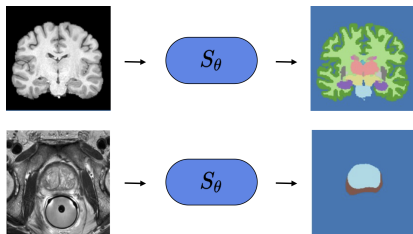


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- Particularly important for MRI due to lack of absolute intensities
- Less of a problem for modalities with absolute intensities, e.g. Hounsfield Unit in CT, however variations can still occur.
- Intensity variations can cause adverse effects for the rest of the processing steps
- Particularly problematic for machine learning methods. Intensity differences between training and test samples is called domain shift. Think of images at night and day.
- Remains an open problem

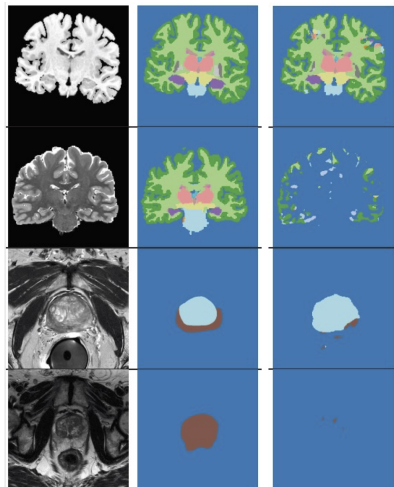
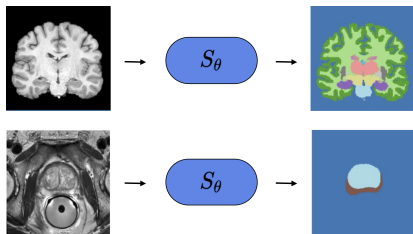
Example



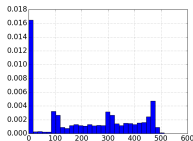
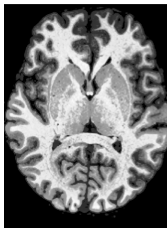
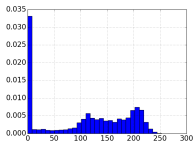
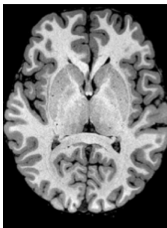
Example



Example



In the ideal case



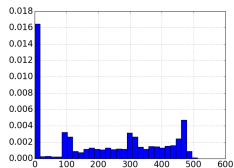
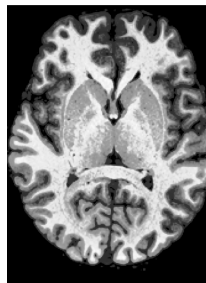
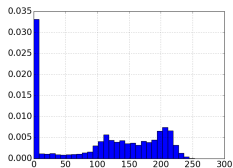
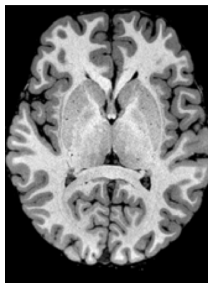
Whenever

- Two images are compared, e.g. registration
- Two groups are compared, e.g. statistical analysis
- A model's parameters are estimated from a set of images, i.e. training, and estimated model is applied on a new set of images, i.e. testing

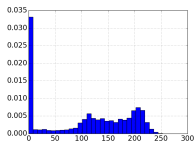
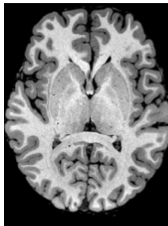
You would like images to have the same intensity "characteristics".

Notion of "characteristics" refer to intensity distribution, including global and local intensity and contrast.

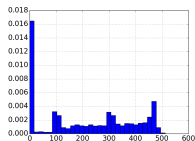
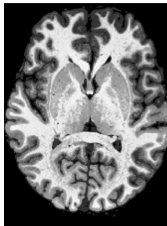
How would you solve this?



Approach I: Min-max



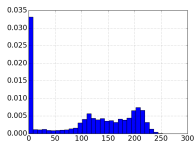
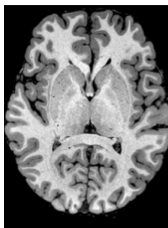
Target Histogram



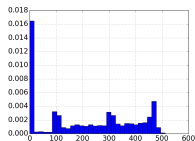
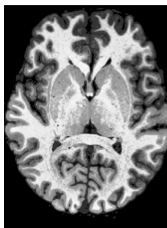
Original Histogram

- Matching the range

Approach I: Min-max



Target Histogram



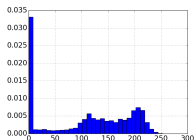
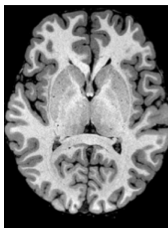
Original Histogram

- Matching the range
- Min-max normalization

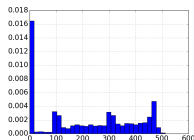
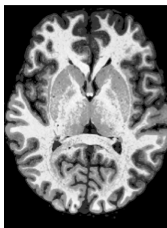
$$f(j) = \frac{j - j_{min}}{j_{max} - j_{min}}(i_{max} - i_{min}) + i_{min}$$

where i and j are the intensities of the two images.

Approach I: Min-max



Target Histogram



Original Histogram

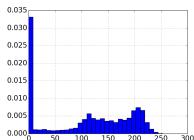
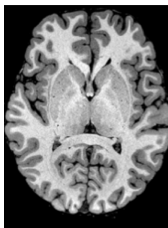
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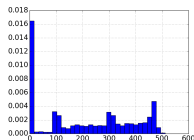
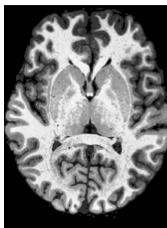
where i and j are the intensities of the two images.

- max and min values are sensitive to outliers
- Often intensities corresponding to $\epsilon\%$ and $1 - \epsilon\%$ of the CDF are used instead, e.g., $\epsilon = 2$

Approach I: Min-max



Target Histogram



Original Histogram

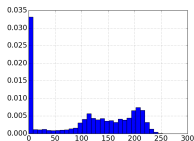
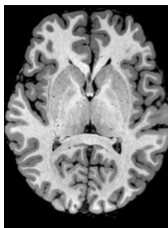
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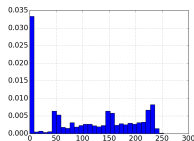
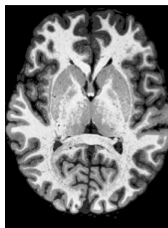
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- Commonly used in machine-learning based methods as a preprocessing step

Approach I: Min-max



Target Histogram



Updated Histogram

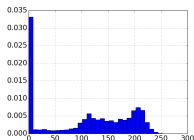
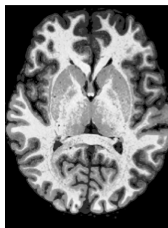
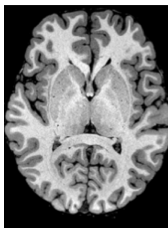
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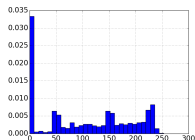
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- max and min values are sensitive to outliers
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What does min-max solve and not solve?



Target Histogram



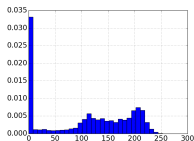
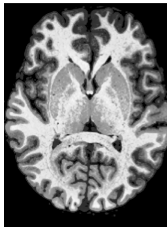
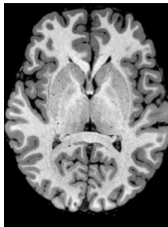
Updated Histogram

■ Min-max normalization

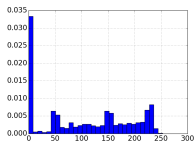
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where i and j are the intensities of the two images.

What does min-max solve and not solve?



Target Histogram



Updated Histogram

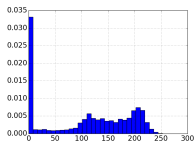
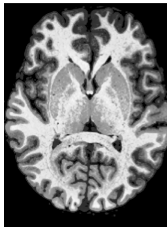
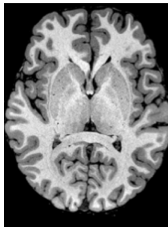
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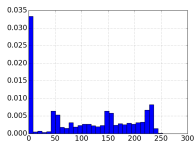
where i and j are the intensities of the two images.

- Can it solve global intensity shifts and multiplicative factors?

What does min-max solve and not solve?



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Updated Histogram

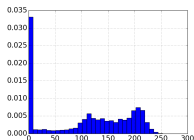
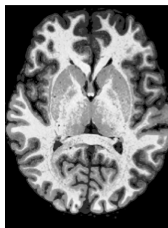
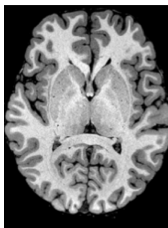
- Min-max normalization

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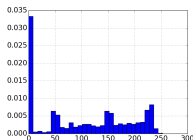
where i and j are the intensities of the two images.

- Can it solve global intensity shifts and multiplicative factors? **Yes.**

What does min-max solve and not solve?



Target Histogram



Updated Histogram

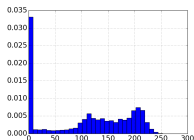
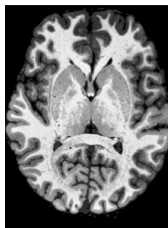
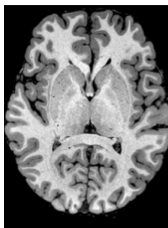
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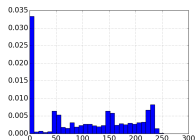
where i and j are the intensities of the two images.

- Can it solve global intensity shifts and multiplicative factors? **Yes.**
- Can it solve contrast differences?

What does min-max solve and not solve?



Target Histogram



Updated Histogram

- Min-max normalization

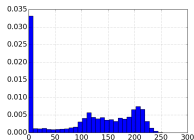
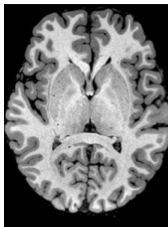
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where i and j are the intensities of the two images.

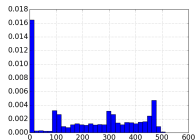
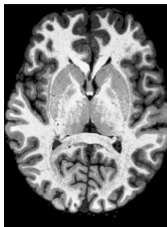
- Can it solve global intensity shifts and multiplicative factors? **Yes.**
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Approach II: Matching mean and standard deviation

- Matching second order statistics:
Mean and standard deviation
normalization



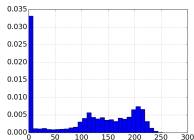
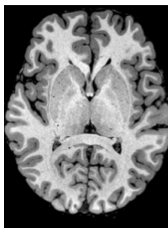
Target Histogram



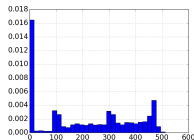
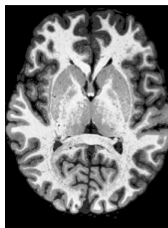
Original Histogram

Approach II: Matching mean and standard deviation

- Matching second order statistics:
Mean and standard deviation
normalization

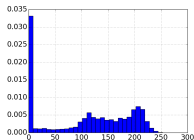
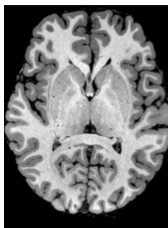


Target Histogram

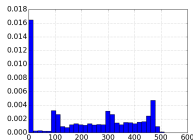
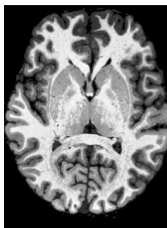


Original Histogram

Approach II: Matching mean and standard deviation



Target Histogram



Original Histogram

- Matching second order statistics:
Mean and standard deviation
normalization

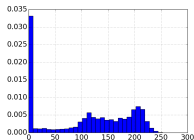
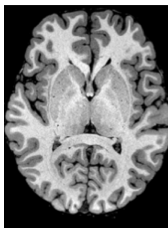
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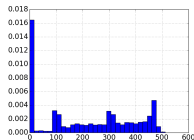
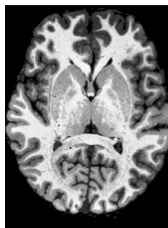
$$\sigma_i^2 = \frac{1}{N-1} \sum_{x=1}^N (i(x) - \mu_i)^2$$

- Less sensitive to outliers but still robust statistics may be used for better behavior, e.g. median, robust variance

Approach II: Matching mean and standard deviation



Target Histogram



Original Histogram

- Matching second order statistics:
Mean and standard deviation
normalization

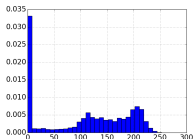
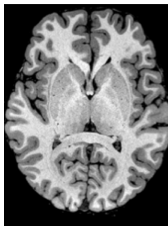
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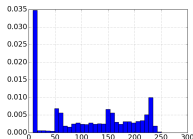
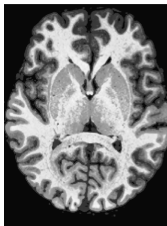
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- Often used in machine-learning based methods

Approach II: Matching mean and standard deviation



Target Histogram



Updated Histogram

- Matching second order statistics:
Mean and standard deviation
normalization

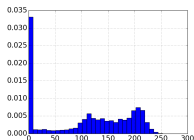
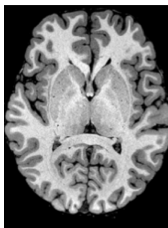
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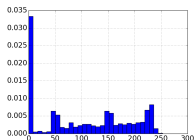
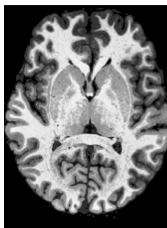
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What does mean and standard deviation matching solve and not solve?



Target Histogram



Updated Histogram

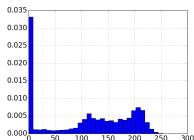
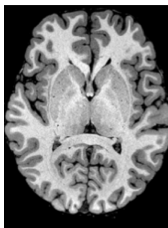
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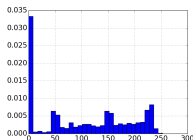
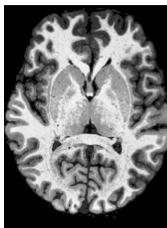
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Target Histogram



Updated Histogram

- Matching second order statistics:
Mean and standard deviation
normalization

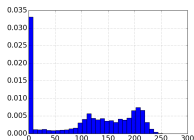
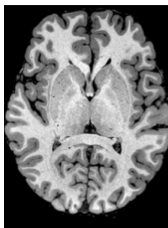
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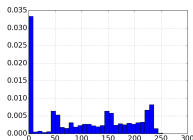
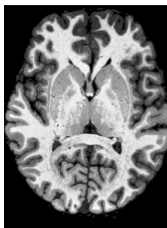
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Target Histogram



Updated Histogram

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Mean and standard deviation
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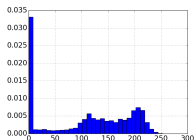
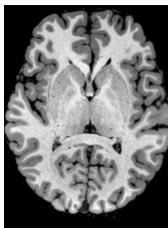
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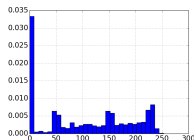
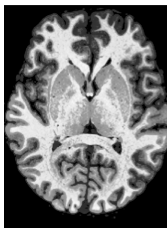
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Target Histogram



Updated Histogram

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Mean and standard deviation
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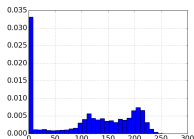
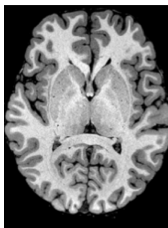
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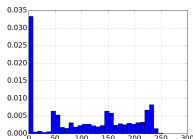
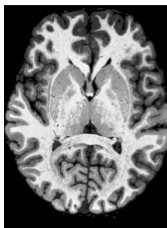
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Question

Min-max matching

$$f(j) = \frac{j - j_{min}}{j_{max} - j_{min}} (i_{max} - i_{min}) + i_{min}$$

Mean - std matching

$$f(j) = (j - \mu_j) \frac{\sigma_i}{\sigma_j} + \mu_i$$

$$\mu_i = \frac{1}{N} \sum_{x=1}^N i(x)$$

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Both min-max and mean-std matching are linear mappings*. True or False?

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Can these transformations change the contrast of an image?

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We need a **non-linear transformations** in the intensity space.

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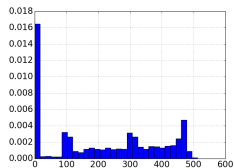
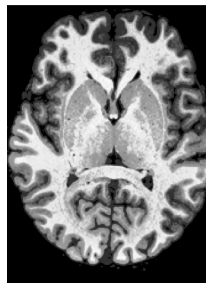
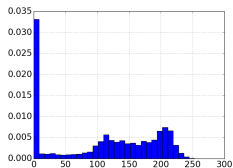
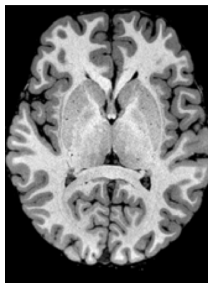
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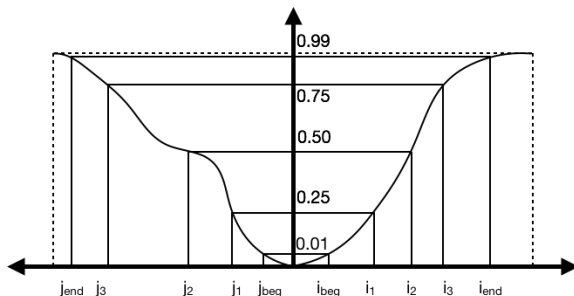
We need a **non-linear transformations** in the intensity space.

(*: linear except the bias or linear in the homogeneous coordinates)

What type of non-linear transformation can you think of?



Approach III: Histogram normalization with landmarks



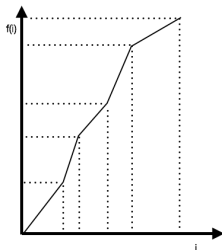
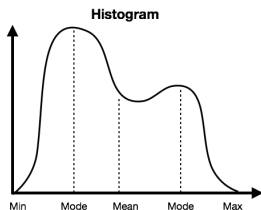
Example: Quartiles and min-max as landmarks

■ Piecewise linear mapping

$$f(j) = \begin{cases} (j - j_{beg}) \frac{i_1 - i_{beg}}{j_1 - j_{beg}} + i_{beg}; & j \leq j_1 \\ (j - j_1) \frac{i_2 - i_1}{j_2 - j_1} + i_1; & j_1 < j \leq j_2 \\ (j - j_2) \frac{i_3 - i_2}{j_3 - j_2} + i_2; & j_2 < j \leq j_3 \\ (j - j_3) \frac{i_{end} - i_3}{j_{end} - j_3} + i_3; & j_3 < j \end{cases}$$

Histogram normalization: Nyul's method

[Nyul, Udupa and Zhang, IEEE TMI 2000]

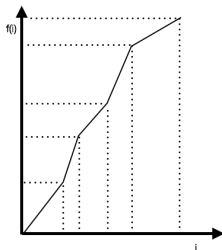
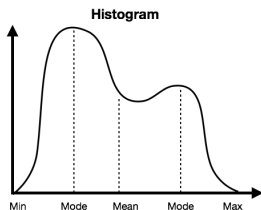


Piecewise linear map

- Matching landmarks of image histograms

Histogram normalization: Nyul's method

[Nyul, Udupa and Zhang, IEEE TMI 2000]

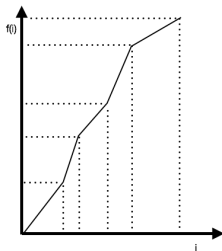
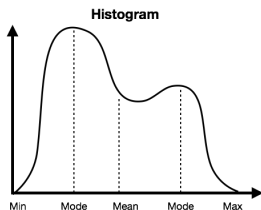


Piecewise linear map

- Matching landmarks of image histograms
- Piece-wise linear map: non-linear mapping

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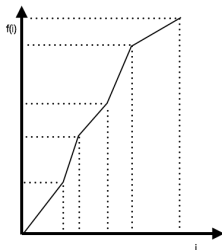
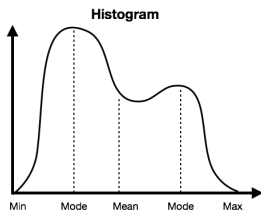


Piecewise linear map

- Matching landmarks of image histograms
- Piece-wise linear map: non-linear mapping
- Different landmarks are possible
 - min and max
 - mean or median
 - quartiles, deciles
 - modes, ...

Histogram normalization: Nyul's method

[Nyul, Udupa and Zhang, IEEE TMI 2000]

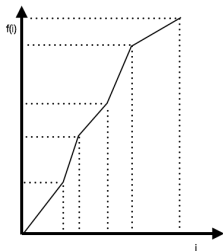
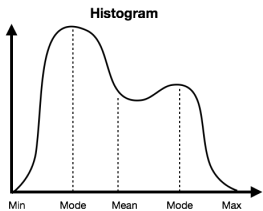


Piecewise linear map

- Matching landmarks of image histograms
- Piece-wise linear map: non-linear mapping
- Different landmarks are possible
 - min and max
 - mean or median
 - quartiles, deciles
 - modes, ...
- Less sensitive to outliers

Histogram normalization: Nyul's method

[Nyul, Udupa and Zhang, IEEE TMI 2000]

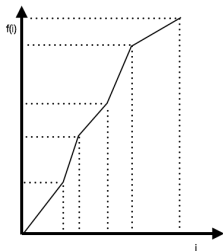
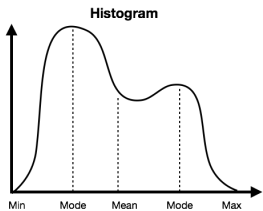


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Histogram normalization: Nyul's method

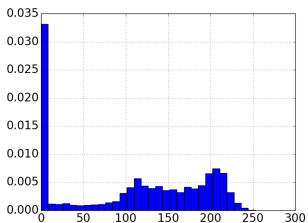
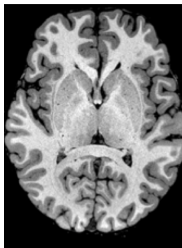
[Nyul, Udupa and Zhang, IEEE TMI 2000]



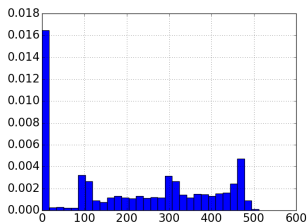
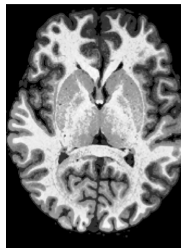
Piecewise linear map

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- Piece-wise linear map: non-linear mapping
- Different landmarks are possible
 - min and max
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 - quartiles, deciles
 - modes, ...
- Less sensitive to outliers
- Widely used
- Population-wide use:
 - Estimate landmark positions from population data as averages
 - Map the new image's landmarks to the averages

Correction results

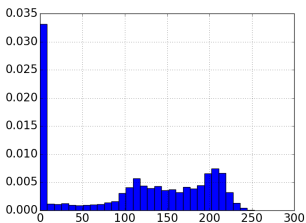
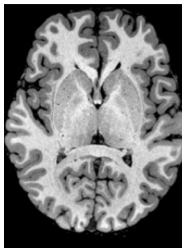


Target Histogram

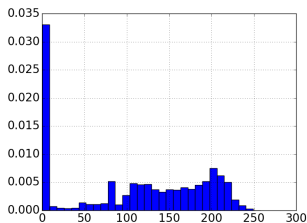
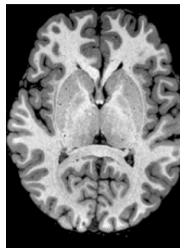


Original Histogram

Correction results

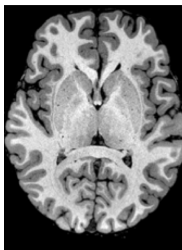


Target Histogram

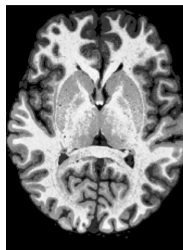


Updated Histogram

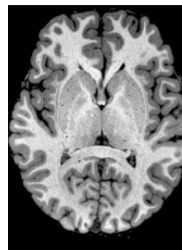
Question



Target image



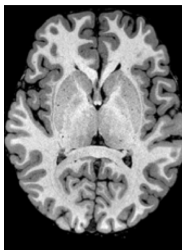
Original image



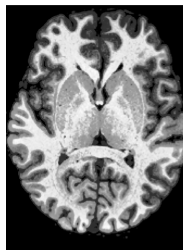
Matched image

Why did the contrast change?

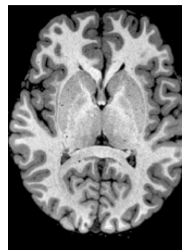
Question



Target image



Original image



Matched image

Why did the contrast change? **Due to the non-linearity of the mapping.**

Analysis

Several things to pay attention to

Analysis

Several things to pay attention to

- Image content

Analysis

Several things to pay attention to

- Image content
 - Histogram matching does not know about image content
 - All pixels are considered independent
 - No spatial information / correlation between neighboring pixels
 - Matching gradients has been proposed

Analysis

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- Image content
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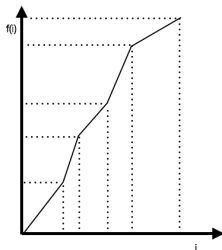
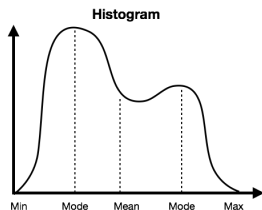
Analysis

Several things to pay attention to

- Image content
 - Histogram matching does not know about image content
 - All pixels are considered independent
 - No spatial information / correlation between neighboring pixels
 - Matching gradients has been proposed
- Object size
 - Differences in object size can be a problem
 - Perfect histogram match would not be preferred in that case
 - Use *mode* matching if object sizes are different

Brilliance of Nyul's method

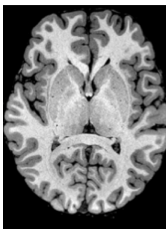
[Nyul, Udupa and Zhang, IEEE TMI 2000]



Piecewise linear map

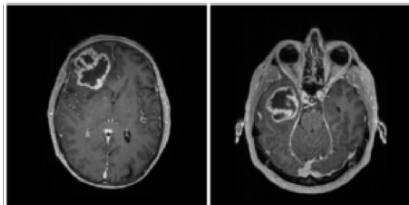
- Matching landmarks of image histograms
- Landmarks: mean, modes, ...
- Matching only the locations of the landmarks **does not make the histograms perfectly equal.**

Analysis - Pathologies



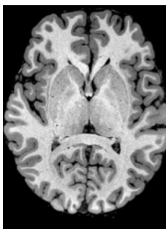
Images of healthy brain

When matching histograms of images with large lesions and healthy brains, should we expect any problems?

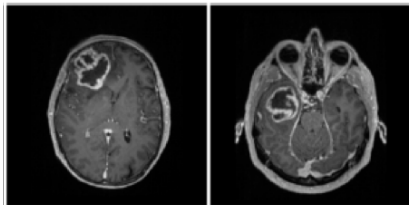


Images with brain tumors

Analysis - Pathologies



Images of healthy brain



Images with brain tumors

When matching histograms of images with large lesions and healthy brains, should we expect any problems?

- Presence of pathologies will change histograms
- Matching pathology bearing images is hard
- *Robust statistics* and *outlier detection* can help

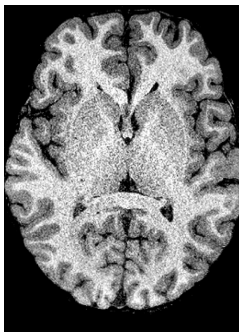
Section 3

Noise suppression

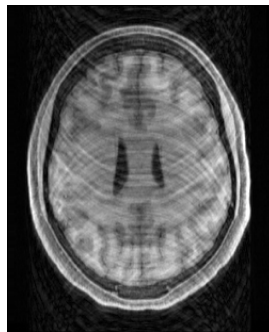
Outline

- Contrast enhancement
- Intensity normalization
- Noise suppression
 - Problem
 - Linear filters
 - Median filters
 - Non-local means
 - Energy-based
- Exercise / Challenge
- Bias correction
- Variations in image and pixel-size
- Additional material
 - Contrast enhancement
 - Linear filters

Problem source



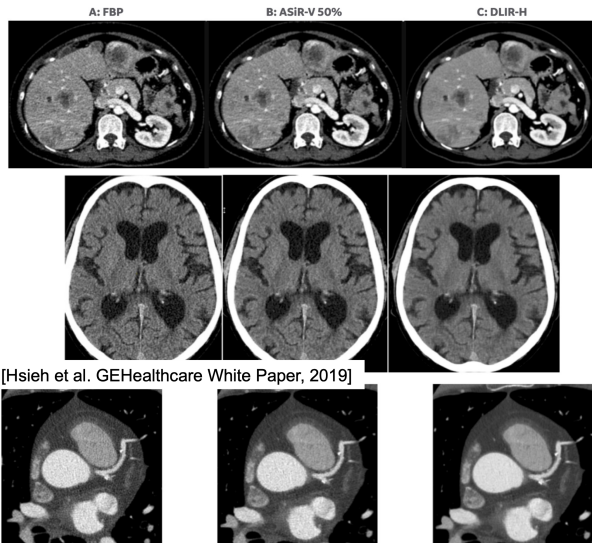
Random noise



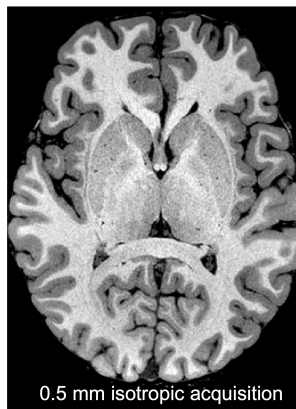
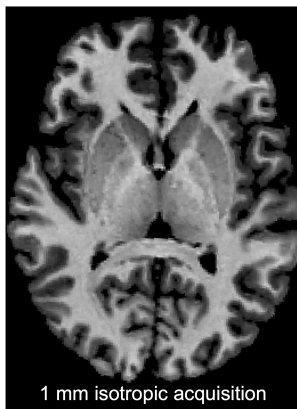
Imaging artifacts

- Imperfections in the acquisition device
- Artifacts due to fast acquisition
- Artifacts due to implants
- ...

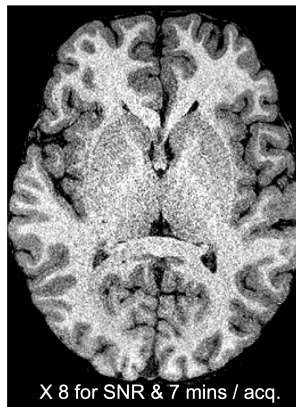
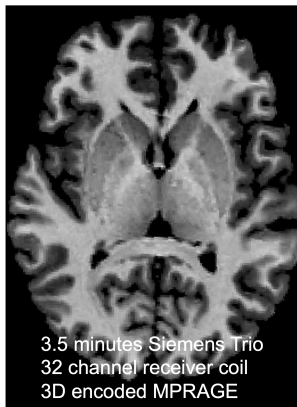
Example I



Example II



Example II



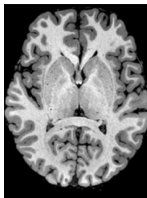
Problem definition

■ Additive noise

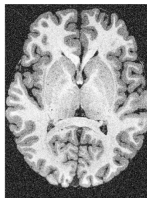
$$J(x) = I(x) + \epsilon(x).$$

$\epsilon(x)$ is often considered to be iid

Example: Gaussian noise ($\epsilon \sim \mathcal{N}(0, \sigma)$)

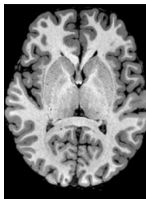


Clean image: I

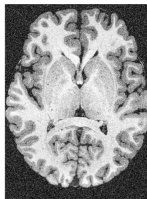


Noisy image: J

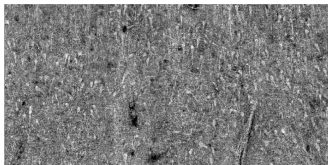
Problem definition



Clean image: I



Noisy image: J



Speckle

■ Additive noise

$$J(x) = I(x) + \epsilon(x).$$

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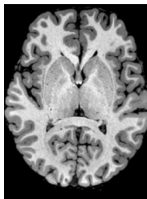
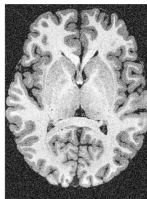
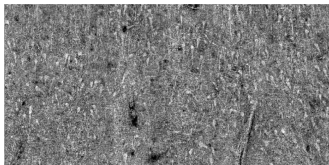
Example: Gaussian noise ($\epsilon \sim \mathcal{N}(0, \sigma)$)

■ Multiplicative noise

$$J(x) = I(x)\epsilon(x)$$

Example: Speckle in ultrasound and optical coherence tomography

Problem definition

Clean image: I Noisy image: J 

Speckle

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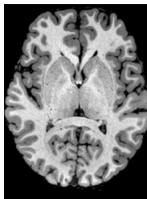
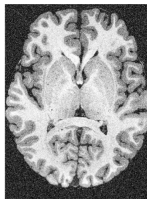
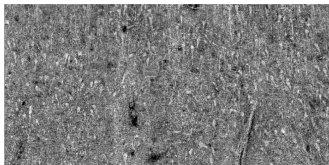
■ More complicated, example: Noise in MR magnitude image

$$J(x) = \sqrt{(I_r(x) + \epsilon)^2 + (I_i(x) + \eta)^2}$$

$\epsilon, \eta \sim \mathcal{N}(0, \sigma)$, Rician noise

[Gudbjartsson and Patz, MRM 1995]

Problem definition

Clean image: I Noisy image: J 

Speckle

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$\epsilon, \eta \sim \mathcal{N}(0, \sigma)$, Rician noise

[Gudbjartsson and Patz, MRM 1995]

■ Given the noisy image J remove noise to get I

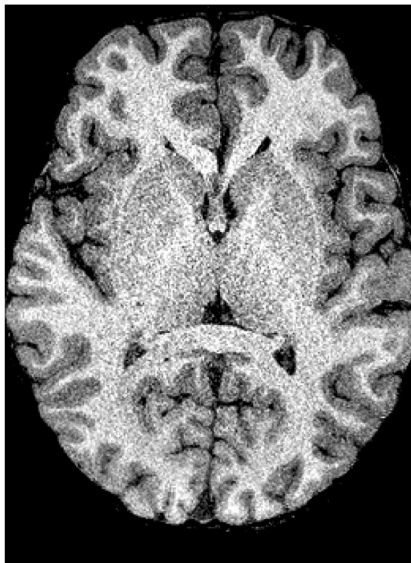
Methods overview

- Generic methods
 - Convolutional / linear
 - Median filtering
 - Anisotropic diffusion
 - Non-local means
 - Optimization based
- Specialized methods: For specific applications and noise models - we will not study them.

Subsection 1

Linear filters

What is the most basic way of tackling this?

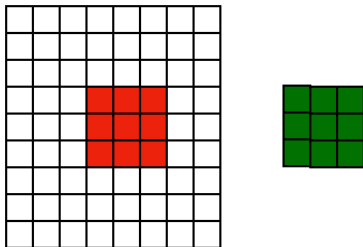


Convolutional / linear filtering

Convolution

$$\begin{aligned}\tilde{I} &= J * S \\ \tilde{I}(x, y, z) &= \int_r \int_p \int_q J(x - r, y - p, z - q) S(r, p, q) dr dp dq \\ \tilde{I}(i, j, k) &= \sum_r \sum_p \sum_q J(i - r, j - p, k - q) S(r, p, q)\end{aligned}$$

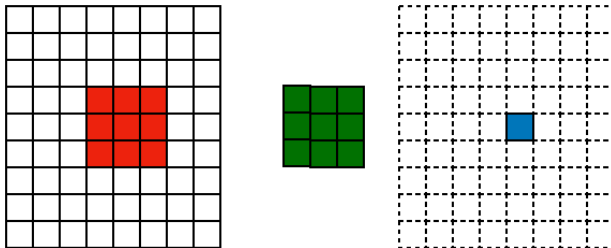
S : structural element (kernel) and $\tilde{I}$ is the denoised image



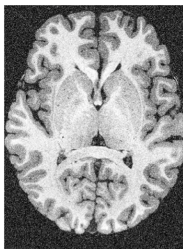
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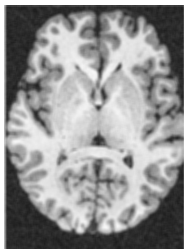
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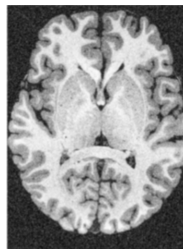
Analysis



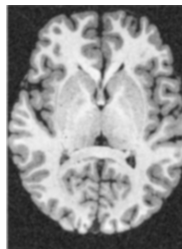
Noisy



Mean with S_7



Binom with S_7



Gauss with $\sigma = 2$

- Extremely efficient to apply
- Easy to implement
- Blurs edges
- Not context aware
- Sensitive to outliers

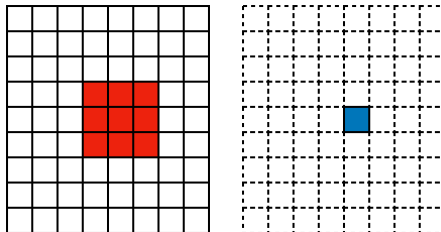
Remark

It is not possible to remove noise and not blur edges at the same time with a linear filter. You need non-linear filtering for this.

Subsection 2

Median filter

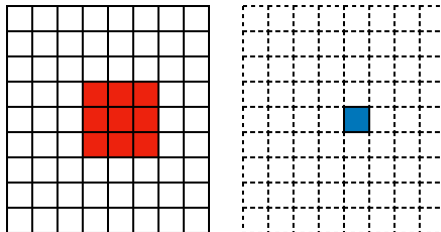
Median filter



- For each location, rank order all neighboring intensities
- Assign the median value as the result

1, 2, 2, 3, 2, 2, 9, 3

Median filter

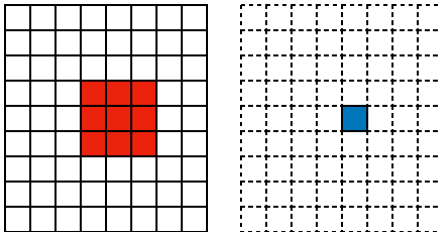


- For each location, rank order all neighboring intensities
- Assign the median value as the result

1, 2, 2, 3, 2, 2, 9, 3

1, 2, 2, 2, 2, 3, 3, 9 $\rightarrow$ 2

Median filter

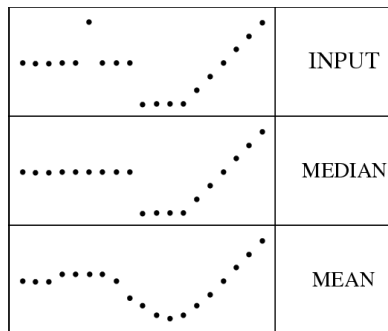


- For each location, rank order all neighboring intensities
- Assign the median value as the result

1, 2, 2, 3, 2, 2, 9, 3
 1, 2, 2, 2, 2, 3, 3, 9 $\rightarrow$ 2

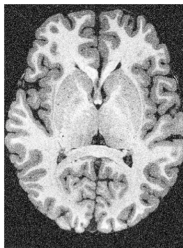
- Sliding window similar to convolution
- Other rank order filters, such as min and max, are defined similarly
- Non-linear filters
- Two rank filters are not commutative: $\min(\text{median}(I)) \neq \text{median}(\min(I))$

Properties



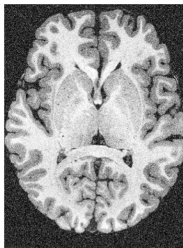
- Robustness to outliers - does not respond to spikes
- Preserves discontinuities, such as edges

Median filtering examples

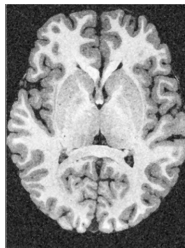


Noisy

Median filtering examples

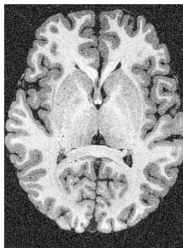


Noisy

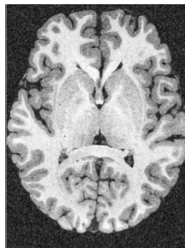


S_3

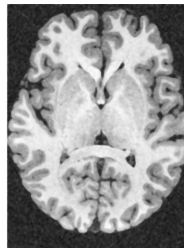
Median filtering examples



Noisy

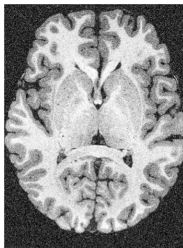


S_3

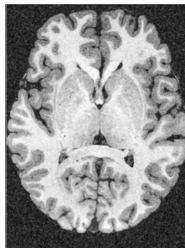


S_5

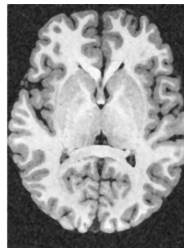
Median filtering examples



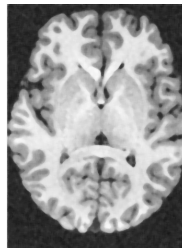
Noisy



S_3

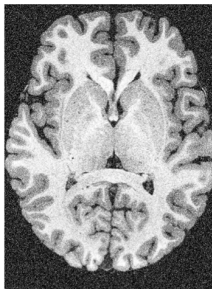


S_5

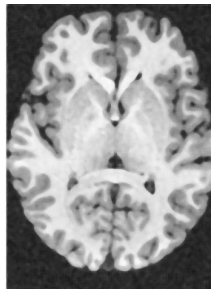


S_7

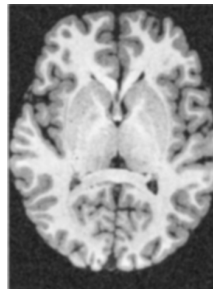
Analysis



Noisy



Median S_7



Gaussian S_7

- Robustness and edge preservation
- Easy and efficient implementation
- Does not introduce new intensities: (i) keeping the intensities integer, (ii) applying median filtering to labels - in a post-processing step - will not introduce new labels and remove islands, e.g., segmentation
- Patchy result
- Image content can change - adding and removing structures

Subsection 3

Non-local means

Non-local means

- [Buades, Coll and Morel, CVPR 2005]
- Very widely used model, very competitive results
- Extension of *bilateral filtering*

Non-local means

- [Buades, Coll and Morel, CVPR 2005]
- Very widely used model, very competitive results
- Extension of *bilateral filtering*
- Most noise suppression methods can be put in the form:

$$I(x) = \sum_{y \in \Omega} w(x, y) J(y) \text{ with } 0 \leq w(x, y) \leq 1 \text{ and } \sum_{y \in \Omega} w(x, y) = 1, \forall x$$

e.g. mean and Gaussian filter. *Weights* $w(x, y)$ changes accordingly.

Non-local means

- [Buades, Coll and Morel, CVPR 2005]
- Very widely used model, very competitive results
- Extension of *bilateral filtering*
- Most noise suppression methods can be put in the form:

$$I(x) = \sum_{y \in \Omega} w(x, y) J(y) \text{ with } 0 \leq w(x, y) \leq 1 \text{ and } \sum_{y \in \Omega} w(x, y) = 1, \forall x$$

e.g. mean and Gaussian filter. *Weights* $w(x, y)$ changes accordingly.

- Non-local means uses $w(x, y)$ to assess similarity between image information at x and y

$$w(x, y) = \frac{1}{Z(x)} \exp \left\{ - \frac{\|J(\mathcal{N}(x)) - J(\mathcal{N}(y))\|_2^2}{h^2} \right\}$$

$Z(x)$ is the normalization constant, h is the degree of filtering and $\mathcal{N}(x)$ neighborhood around x .

Non-local means - intuition



Taken from [Buades et al. CVPR 2005]

$$I(x) = \sum_{y \in \Omega} w(x, y) J(y)$$

$$w(x, y) = \frac{1}{Z(x)} \exp \left\{ - \frac{\|J(\mathcal{N}(x)) - J(\mathcal{N}(y))\|_2^2}{h^2} \right\}$$

- While smoothing point x use areas with similar image content
- If noise is random then averaging similar areas should get rid of it

Non-local means - intuition



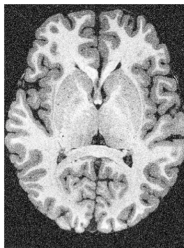
Taken from [Buades et al. CVPR 2005]

$$I(x) = \sum_{y \in \Omega} w(x, y) J(y)$$

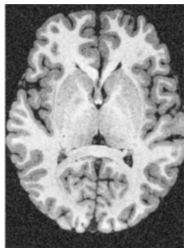
$$w(x, y) = \frac{1}{Z(x)} \exp \left\{ - \frac{\|J(\mathcal{N}(x)) - J(\mathcal{N}(y))\|_2^2}{h^2} \right\}$$

- While smoothing point x use areas with similar image content
- If noise is random then averaging similar areas should get rid of it
- Uses *redundancy* in the images, i.e. local structures repeat

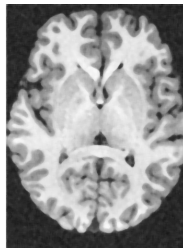
Results



Noisy

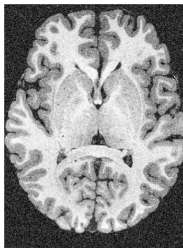


Binom with S_7

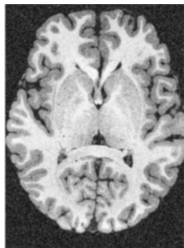


Median with S_7

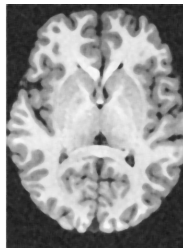
Results



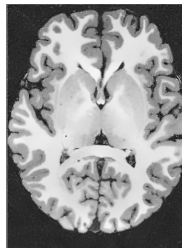
Noisy



Binom with S_7

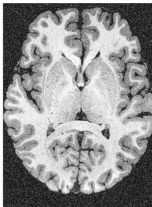


Median with S_7



NLM $h = 15$

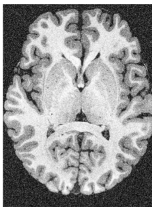
Properties



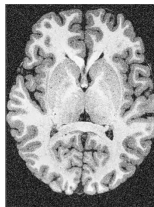
Noisy

- Increasing h leads to more smoothing

Properties



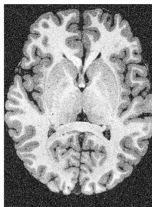
Noisy



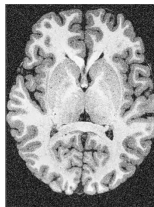
$h = 5$

- Increasing h leads to more smoothing

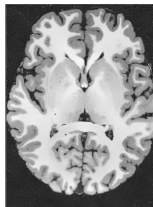
Properties



Noisy



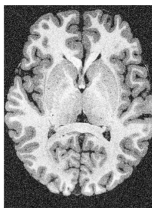
$h = 5$



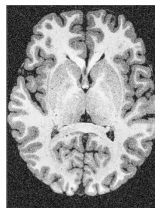
$h = 15$

- Increasing h leads to more smoothing

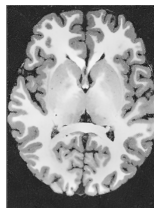
Properties



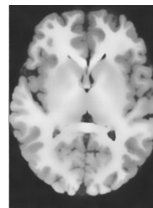
Noisy



$h = 5$



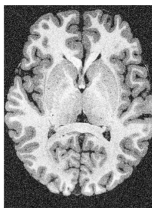
$h = 15$



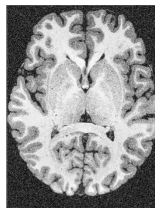
$h = 50$

- Increasing h leads to more smoothing

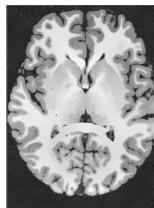
Properties



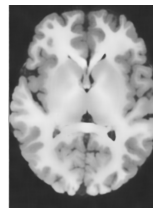
Noisy



$h = 5$



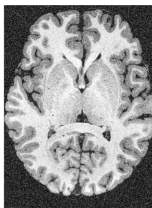
$h = 15$



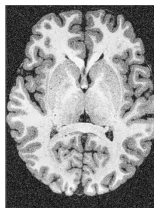
$h = 50$

- Increasing h leads to more smoothing
- Model parameters
 - h - depends on the noise level
 - size of $\mathcal{N}(x)$
 - size of $W(x)$

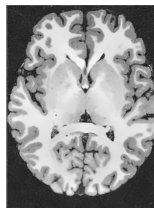
Properties



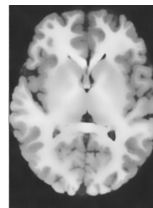
Noisy



$h = 5$



$h = 15$

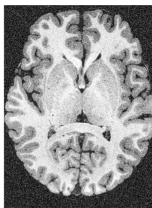


$h = 50$

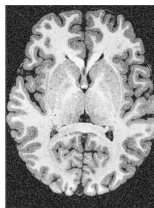
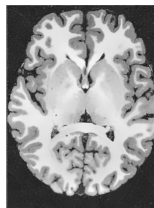
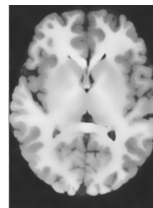
- Increasing h leads to more smoothing
- Model parameters
 - h - depends on the noise level
 - size of $\mathcal{N}(x)$
 - size of $W(x)$
- The averaging often happens in a small *search-window*

$$I(x) = \sum_{y \in W(x)} w(x, y) J(y)$$

Properties



Noisy

 $h = 5$  $h = 15$  $h = 50$

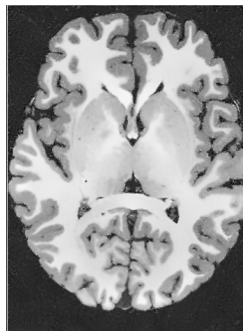
- Increasing h leads to more smoothing
- Model parameters
 - h - depends on the noise level
 - size of $\mathcal{N}(x)$
 - size of $W(x)$
- The averaging often happens in a small *search-window*

$$I(x) = \sum_{y \in W(x)} w(x, y) J(y)$$

- Size of $\mathcal{N}(x)$ matters:
 - $\mathcal{N}(x) = \Omega$ leads to mean filtering
 - $\mathcal{N}(x) = \{x\}$ leads to Yaroslavsky's filter, a special case of bilateral filtering

Analysis

- Higher noise suppression
- Better preservation of edges
- May create artifacts
- Important to estimate noise level - different algorithms to do it
 - [Coupe et al. Medical Image Analysis 2010]
 - ...
- Extension to MRI and diffusion MRI
 - [Manjon et al. Medical Image Analysis 2008]
 - [Wiest-Daessle et al. MICCAI 2008]
 - ...
- Currently the state-of-the-art filtering technique that is not based on machine learning.



Comparison

Convolutional Filters

Median Filters

Non-local means

Comparison

Convolutional Filters

- Efficient to apply and easy to implement
- Linear: easy to analyze mathematically
- Blurs edges
- Not context aware
- Sensitive to outliers

Median Filters

- Efficient to apply and easy to implement
- Robust to outliers
- Edge preservation
- No new intensities introduced
- Patchy results
- Image content may change

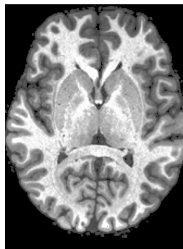
Non-local means

- Higher noise suppression
- Better edge preservation
- May create artifacts
- Not as efficient to apply as the others

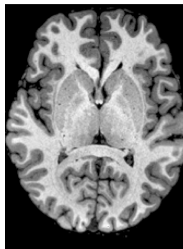
Section 4

Summary and Exercise

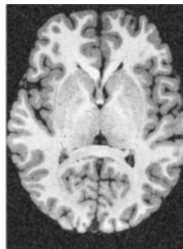
Summary



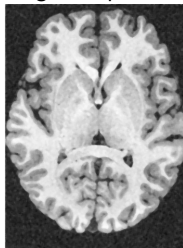
Histogram equalization



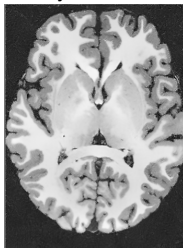
Intensity normalization



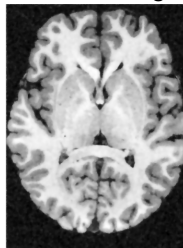
Linear filtering



Median filtering

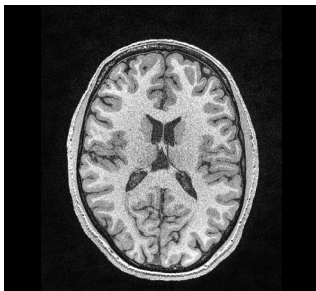


Non-local means

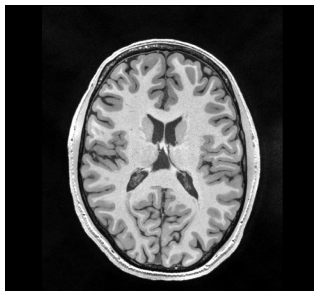


TV with $\lambda = 0.3$

Noise suppression challenge



Single acquisition



Average of 8

Challenge

High-resolution MRI is hard to acquire. One way is to acquire many images of the same subject and average them. 8 or so images yield good SNR but this leads to long acquisitions. Can you reduce this time with noise suppression?

More details in the moodle platform.

Outline - Next week

- Contrast enhancement
- Intensity normalization
- Noise suppression
- Exercise / Challenge
- Bias correction
- Variations in image and pixel-size
- Discussions about the exercise

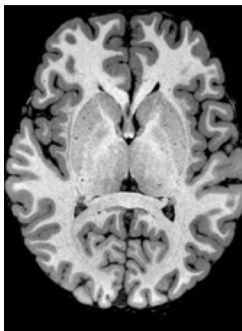
Section 5

Additional Materials

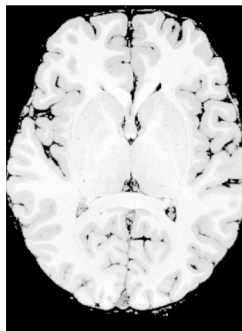
Subsection 1

Contrast enhancement

Problem description



Original image



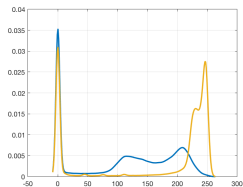
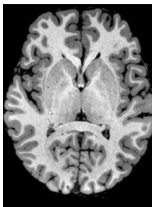
Observation with bad contrast

Under-"exposure" or over-"exposure"

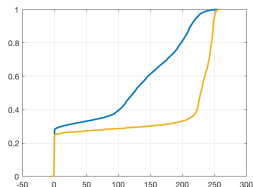
How to enhance the contrast of the observation?

Problem description in terms of histograms

In a nice image intensities are well spread across the range.



Histograms

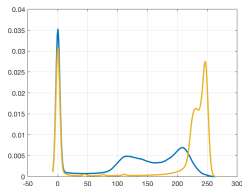
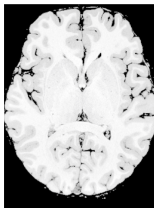


Cumulative histograms

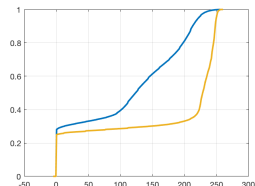
Figure: Blue curves are from the image with good contrast.

Problem description in terms of histograms

In an image with bad contrast, interesting parts of the image are squeezed in a small intensity range.



Histograms

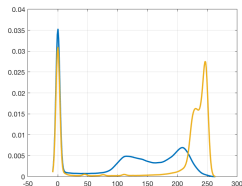
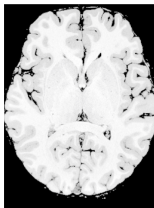


Cumulative histograms

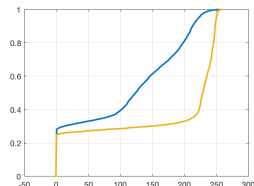
Figure: Yellow curves are from the image with bad contrast.

Problem description in terms of histograms

In an image with bad contrast, interesting parts of the image are squeezed in a small intensity range.



Histograms



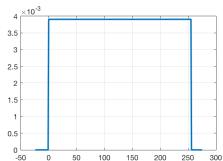
Cumulative histograms

Figure: Yellow curves are from the image with bad contrast.

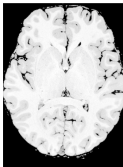
The goal is to **distribute the intensities to a larger range**

Flat histogram

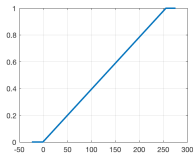
Approach: **Flatten** the histogram to cover as much intensity region as possible



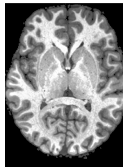
Flat Histogram



Initial image



CDF of a Flat Histogram



Flattened image

Histogram equalization principle

Strategy: Find a **monotonically increasing** intensity mapping that will redistribute intensities to flatten the histogram:

$$\forall i \in [0, i_{max}], i \rightarrow f(i), \text{ such that } \forall i \geq j \text{ we have } f(i) \geq f(j)$$

Histogram equalization principle

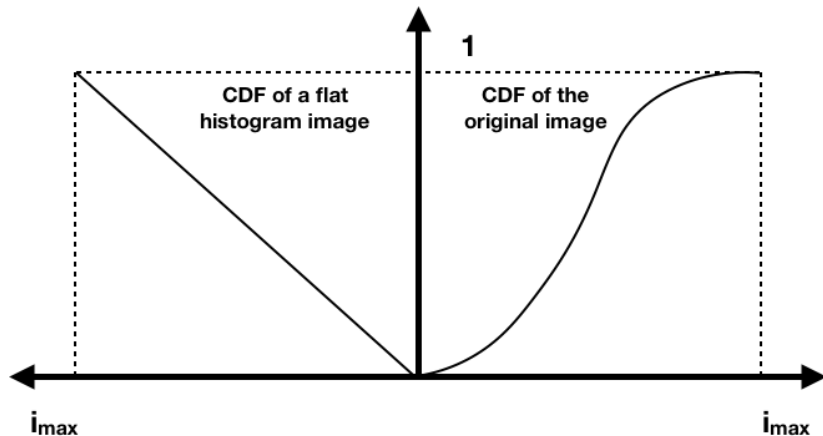
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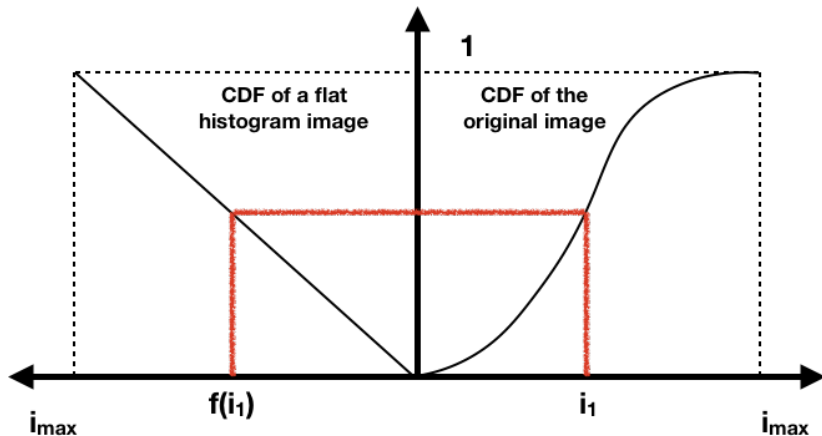
A simple mapping that would achieve this in the continuous domain is the cumulative density distribution itself

$$f(i) = i_{max}P(i) = i_{max} \int_0^i p(j)dj$$

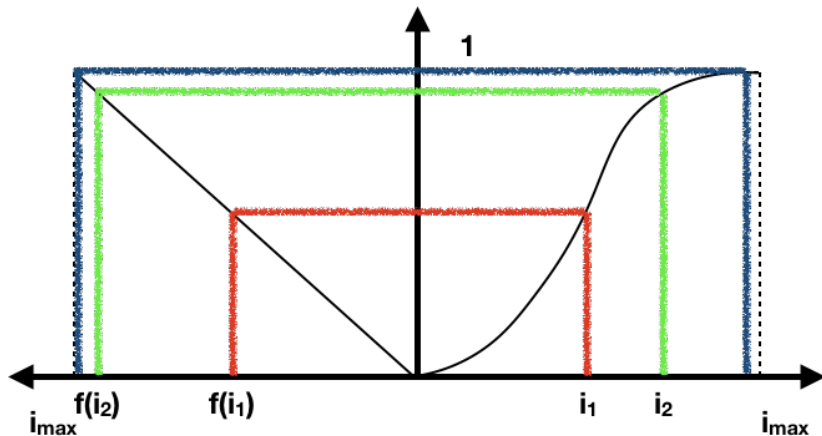
Histogram equalization pictorially



Histogram equalization pictorially



Histogram equalization pictorially



Histogram equalization principle

Strategy: Find a monotonically increasing intensity mapping that will redistribute intensities to flatten the histogram

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Histogram equalization principle

Strategy: Find a monotonically increasing intensity mapping that will redistribute intensities to flatten the histogram

$$\forall i \in [0, i_{\max}], i \rightarrow f(i), \text{ such that } \forall i \geq j \text{ we have } f(i) \geq f(j)$$

A simple mapping that would achieve this in the continuous domain is the cumulative density distribution itself

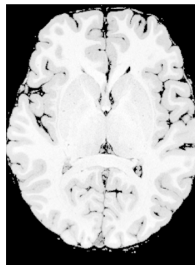
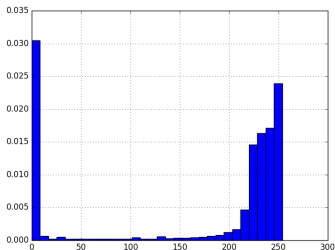
$$f(i) = i_{\max} P(i) = i_{\max} \int_0^i p(j) dj$$

It is easy to see this mapping flattens the distribution

$$p(f(i)) = p(i) \frac{di}{df(i)} = p(i) \frac{1}{p(i)} \frac{1}{i_{\max}} = \frac{1}{i_{\max}}$$

Histogram equalization algorithmically

In the quantized image we will get discrete histogram:



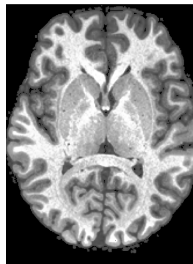
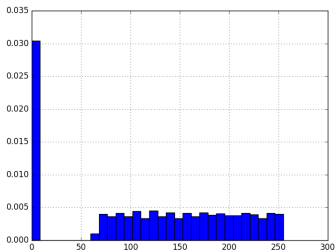
The mapping is:

$$f(i) = i_{max} \sum_{j=0}^i h_j(j), \quad i_{max} : \text{Maximum intensity in the original image.}$$

$h_j(j)$ is the histogram value at intensity j

Histogram equalization result

Corrected histogram and the image:

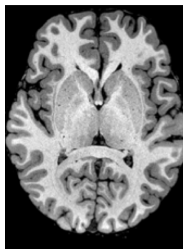


The mapping is then:

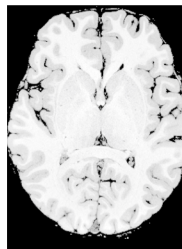
$$f(i) = i_{max} \sum_{j=0}^i h_I(j), \quad i_{max} : \text{Maximum intensity in the original image.}$$

$h_I(j)$ is the histogram value at intensity j

Question



Original image



Observation with bad contrast

Histogram equalization improves contrast for under and over exposed images.
Can it improve the image if intensities of two different tissue types overlap?

Subsection 2

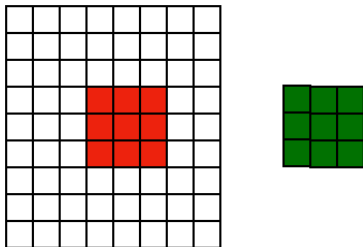
Linear filters

Convolutional / linear filtering

Convolution

$$\begin{aligned}\tilde{I} &= J * S \\ \tilde{I}(x, y, z) &= \int_r \int_p \int_q J(x - r, y - p, z - q) S(r, p, q) dr dp dq \\ \tilde{I}(i, j, k) &= \sum_r \sum_p \sum_q J(i - r, j - p, k - q) S(r, p, q)\end{aligned}$$

S : structural element (kernel) and $\tilde{I}$ is the denoised image

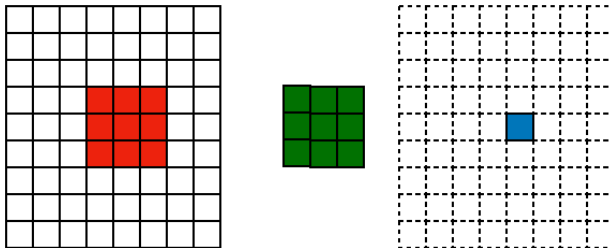


Convolutional / linear filtering

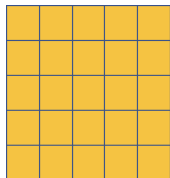
Convolution

$$\begin{aligned}\tilde{I} &= J * S \\ \tilde{I}(x, y, z) &= \int_r \int_p \int_q J(x - r, y - p, z - q) S(r, p, q) dr dp dq \\ \tilde{I}(i, j, k) &= \sum_r \sum_p \sum_q J(i - r, j - p, k - q) S(r, p, q)\end{aligned}$$

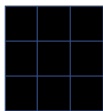
S : structural element (kernel) and $\tilde{I}$ is the denoised image



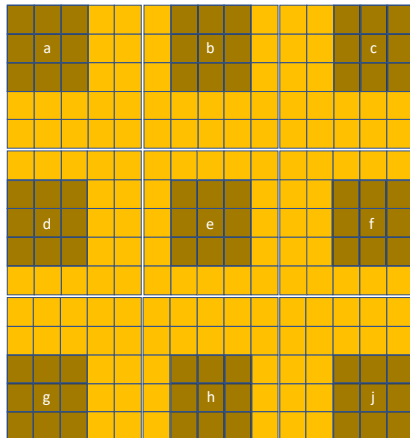
Convolutional / linear filtering - graphically



Image



Kernel



a	b	c
d	e	f
g	h	j

Commonly used convolutional filters for noise suppression: Mean filtering

Mean filtering

- average in a neighborhood

$$S_3 = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

- larger kernels similar, e.g. S_5 and S_7
- larger kernels smooths more

Commonly used convolutional filters for noise suppression: Mean filtering

Mean filtering

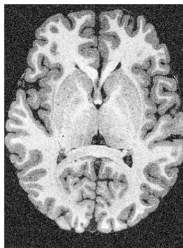
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- larger kernels smooths more
- separable: very efficient to implement

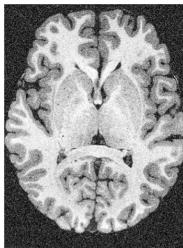
$$S_3 = \frac{1}{9} [1, 1, 1] * \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Mean filtering examples

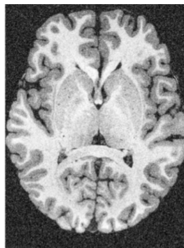


Noisy

Mean filtering examples

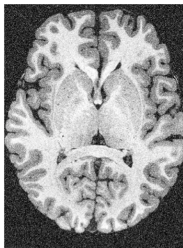


Noisy

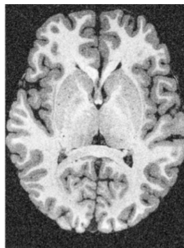


S_3

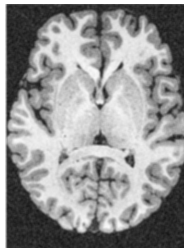
Mean filtering examples



Noisy

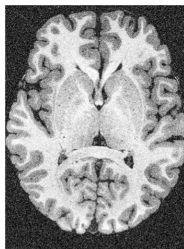


S_3

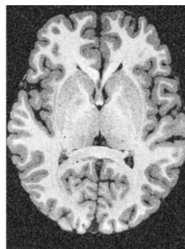


S_5

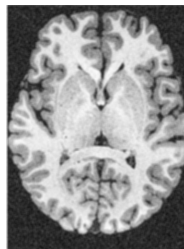
Mean filtering examples



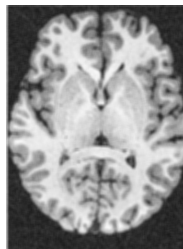
Noisy



S_3



S_5



S_7

Commonly used convolutional filters for noise suppression: Binomial filtering

Binomial filtering

- iterative convolutions of $[1, 1]$
- integer filters

$$S_3 = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Commonly used convolutional filters for noise suppression: Binomial filtering

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Commonly used convolutional filters for noise suppression: Binomial filtering

Binomial filtering

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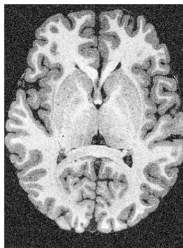
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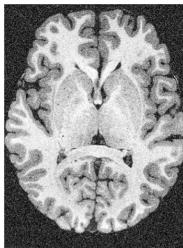
- better properties than averaging filtering in the frequency domain - we will not see this further.

Binomial filtering examples

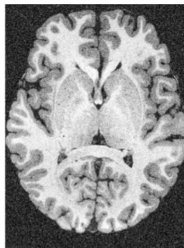


Noisy

Binomial filtering examples

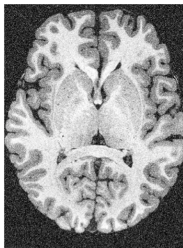


Noisy

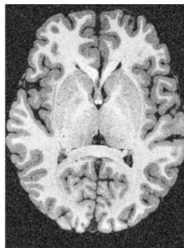


S_3

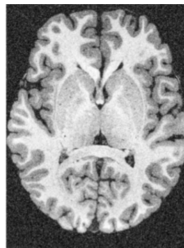
Binomial filtering examples



Noisy

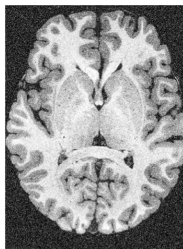


S_3

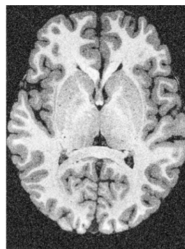


S_5

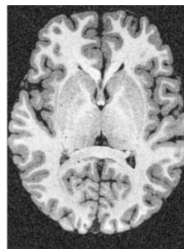
Binomial filtering examples



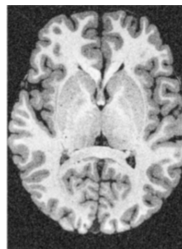
Noisy



S_3



S_5



S_7

Commonly used convolutional filters for noise suppression: Gaussian filtering

Gaussian filtering

- construction by Gaussian kernel centered at the center of the structural element, e.g.

$$S_3(i, j) = \frac{1}{C} \exp \left(-\frac{(i-1)^2 + (j-1)^2}{2\sigma^2} \right), \quad C : \text{Normalizing constant}$$

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- has infinite support
- finite structural approximates the Gaussian kernel

Commonly used convolutional filters for noise suppression: Gaussian filtering

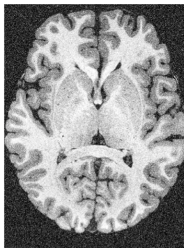
Gaussian filtering

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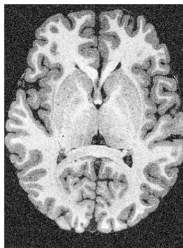
- has infinite support
- finite structural approximates the Gaussian kernel
- structural element size depends on σ , e.g. large σ needs larger size for better approximation
- larger σ filters smooth more

Gaussian filtering examples

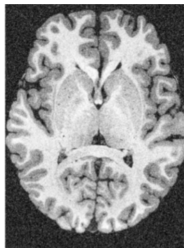


Noisy

Gaussian filtering examples

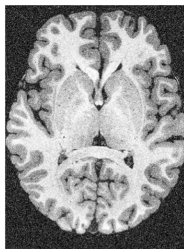


Noisy

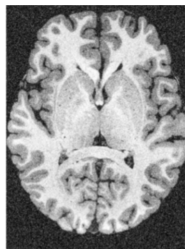


$\sigma = 1$

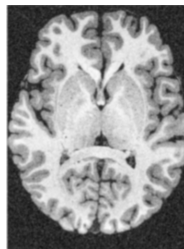
Gaussian filtering examples



Noisy

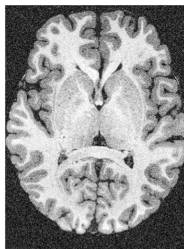


$\sigma = 1$

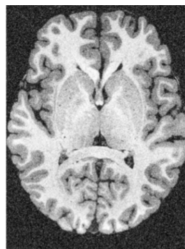


$\sigma = 1.5$

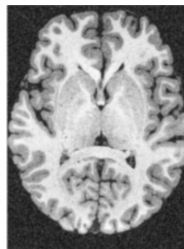
Gaussian filtering examples



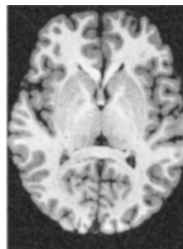
Noisy



$\sigma = 1$



$\sigma = 1.5$



$\sigma = 2$